MEETING AGENDA

Basel & Ziad × Dr. Mohamed Ahmed Radwan

Purpose: present three credible graduation-project directions, demonstrate research readiness, and agree on the best-scoped direction for a seven-person team over ten months.

SUCCESS CRITERION
Leave with a clear next step: a preferred research direction, a focused research question, an initial literature-review task, and a follow-up date—not necessarily a final project approval during the first meeting.

1. Proposed 30-minute agenda

TIME	TOPIC	LEAD	WHAT TO ACHIEVE
0–3 min	**Introductions & motivation**	Basel	Introduce both students; explain why his DFE and power-management background is relevant.
3–7 min	**Team & execution capability**	Ziad	Explain the seven-person split: 2 digital ASIC, 2 verification, 2 analog, 1 leader/integration.
7–12 min	**Existing work: 224 Gb/s MATLAB**	Ziad, then Basel	Show what works, the reference paper, current SER/BER status, and what still needs validation. Do not overclaim.
12–19 min	**Present the three options**	Basel	Summarize AetherRV, the 400 Gb/s ADC receiver, and Idea C; emphasize that you seek his guidance on scope and novelty.
19–25 min	**Technical discussion**	Both	Ask about research gap, DFE architecture, supply modeling, analog circuit feasibility, and the minimum publishable contribution.
25–30 min	**Agree on next steps**	Basel	Confirm preferred direction, first deliverable, literature task, tool/PDK constraints, and follow-up meeting.

2. Speaking roles: who says what

PERSON	RESPONSIBILITY	KEY MESSAGE
Basel	Project framing, architecture, novelty, controller idea, integration	“I can lead the digital architecture and adaptation algorithm, coordinate the team, and turn the research question into measurable experiments.”
Ziad	MATLAB progress, reference comparison, SER/BER evidence, current limitations	“We have a starting point at 224 Gb/s; we want to validate it rigorously and build the next contribution on top of it.”
Both	Listen, take notes, clarify feasibility and supervision expectations	Show that you are open to narrowing the scope based on his advice; do not imply the novelty is already proven.

MEETING SCRIPT & TECHNICAL CHECKLIST

3. Opening statement (Basel)

“Thank you, Dr. Mohamed, for meeting with us. We are Basel and Ziad, and we are exploring a graduation project that can produce a rigorous hardware research contribution. We prepared three candidate directions and would value your help choosing a technically strong and feasible research question that matches your experience in reconfigurable DFE architectures and power-management ICs.”

4. Three ideas — keep each to about 45 seconds

IDEA	ONE-SENTENCE SUMMARY	WHAT TO ASK HIM
A — AetherRV	A RISC-V SoC with an INT8 AI accelerator and supply-droop-aware activity/clock control.	Is this a good fit if we focus on power-integrity-aware computing, and should we add a digital-VLSI co-supervisor?
B — 400 Gb/s ADC receiver	Extend the existing 224 Gb/s MATLAB receiver toward 400 Gb/s, with SER/BER validation and analog/DSP co-design.	What is the most realistic progression from our baseline to a defensible 400 Gb/s result?
C — Supply-aware energy-proportional PAM-N receiver	Coordinate reconfigurable multi-stage DFE settings with supply-aware operating decisions to meet a BER target with lower energy.	Is the joint DFE/power-awareness contribution novel enough, and which part should be the main research contribution?

5. Questions to prioritize

- **Research gap:** What precise gap in existing DFE and power-management research should we investigate?
- **Feasibility:** What can we realistically complete in ten months with our available PDK, tools, and seven-person team?
- **Method:** Which reference paper, channel model, DFE configuration, and BER/SER target should anchor the baseline?
- **Analog scope:** Which one circuit block is worth implementing at transistor level, if any?
- **Publication:** What minimum experimental evidence would make the contribution credible for a conference or journal submission?
- **Supervision:** Would you be interested in supervising this direction, and what meeting cadence and deliverables would you expect?

6. What to bring / show

- **PDF:** the three-idea proposal; use it as a discussion aid, not a script to read aloud.
- **MATLAB:** a short demo or plots showing the current 224 Gb/s receiver and the exact conditions used.
- **Reference:** the paper used for the baseline, plus a one-line summary of what is reproduced and what remains incomplete.
- **Team:** one-slide skills split (2 digital ASIC, 2 verification, 2 analog, Basel as leader/integration).
- **Constraints:** ten-month timeline, available 22 nm FD-SOI PDK/tools, and what has or has not been confirmed.

DECISION NOTES & FOLLOW-UP

7. Fill this in during the meeting

ITEM	NOTES / AGREED ACTION
Preferred project direction	_____
Research question / novelty to test	_____
First deliverable (next 2–3 weeks)	_____
Literature / reference papers to review	_____
Required tools / PDK / access	_____
Analog circuit block (if any)	_____
Supervisor/co-supervisor arrangement	_____
Next meeting / follow-up date	_____

8. Closing statement

“Thank you for your feedback. We would be happy to narrow the scope around the research direction you believe has the strongest technical contribution. We can send you the MATLAB repository, a short literature review, and a revised project plan after the meeting.”

REMEMBER

Your goal is to show research readiness: a real starting point, clear roles, honest limitations, and willingness to refine the novelty with his guidance. Do not oversell results or claim the idea is already novel.